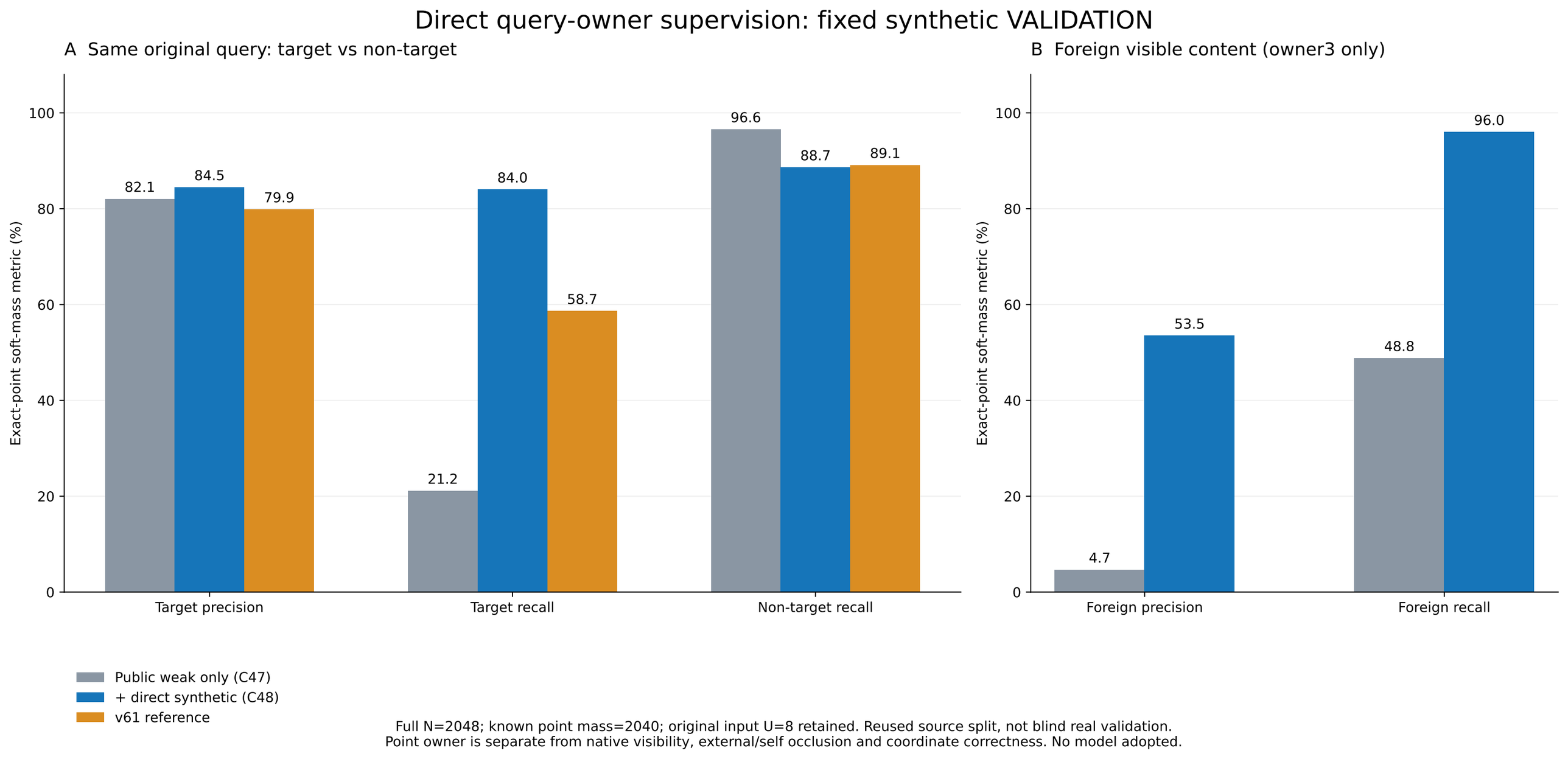
Direct query-owner supervision: fixed synthetic VALIDATION
A Same original query: target vs non-target
B Foreign visible content (owner3 only)
Exact-point soft-mass metric (%)
0
20
40
60
80
100
82.1
84.5
79.9
21.2
84.0
58.7
96.6
88.7
89.1
4.7
53.5
48.8
96.0
Target precision
Target recall
Non-target recall
Foreign precision
Foreign recall
Public weak only (C47)
+ direct synthetic (C48)
v61 reference
Full N=2048; known point mass=2040; original input U=8 retained. Reused source split, not blind real validation.
Point owner is separate from native visibility, external/self occlusion and coordinate correctness. No model adopted.